16. September Week 2: Progress Review and Next-Week Work Plan

16.1 Progress Achieved in Week 1

The first September stage has established the stochastic-process foundation required for the Black–Scholes model.

The completed work now includes

- generation and statistical verification of standard normal random variables;
- construction of Brownian increments;
- simulation of complete Brownian-motion paths;
- numerical verification of

$$E[\Delta W] = 0$$

and

$$\mathrm{Var}(\Delta W) = \Delta t;$$

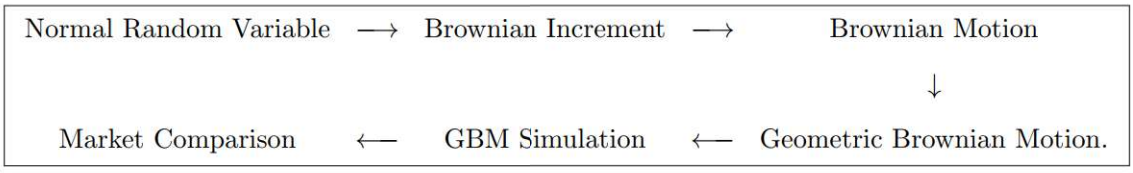
- direct verification of the scaling law

$$\mathrm{SD}(\Delta W) \propto \sqrt{\Delta t};$$

- examination of independence and stationarity of Brownian increments;
- numerical investigation of quadratic variation;
- implementation and validation of geometric Brownian motion;
- investigation of the separate roles of drift and volatility;
- calibration of GBM using NIFTY 50 data; and
- direct comparison of empirical market returns with GBM-generated returns.

16.2 Position Reached After Week 1

The mathematical progression completed so far can be summarized as



The important point is that the stochastic process has now been examined both mathematically and computationally.

The next stage is therefore not to perform further GBM simulations, but to ask:

> If the underlying asset follows GBM,
> what stochastic differential equation is satisfied
> by a derivative whose value is $V(S,t)$?

This question leads directly to Itô's lemma and subsequently to the Black–Scholes equation.

16.3 The Critical Bridge: Quadratic Variation

Week 1 established numerically that Brownian motion behaves fundamentally differently from an ordinary differentiable function.

For an ordinary smooth increment,

$$(\Delta t)^2$$

becomes negligible compared with

$$\Delta t.$$

For Brownian motion,

$$\Delta W \sim \sqrt{\Delta t}.$$

Therefore,

$$(\Delta W)^2 \sim \Delta t.$$

Consequently, the second-order stochastic term cannot be discarded.

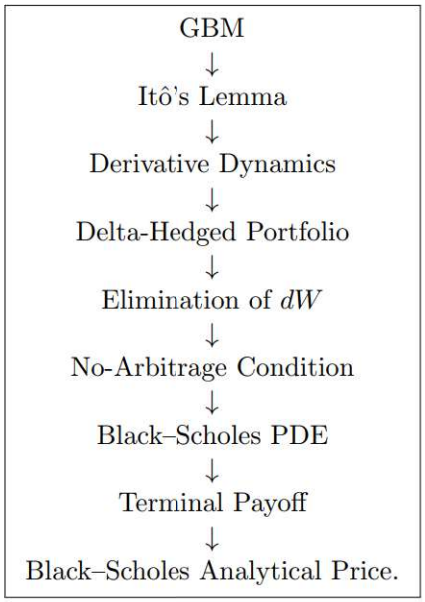
The fundamental stochastic-calculus rules are

$$\boxed{(dW)^2 = dt, \qquad dW\,dt = 0, \qquad (dt)^2 = 0.}$$

This is the mathematical reason why ordinary differential calculus is insufficient for functions of Brownian motion.

16.4 Week 2 Central Objective

The principal objective for Week 2 is to understand and reproduce the complete mathematical chain



The derivation is the principal task of the week.

Programming should support the mathematics rather than replace it.

16.5 Task 1: From Ordinary Taylor Expansion to Itô's Lemma

Begin with a function

$$V = V(S, t).$$

For small changes in S and t, the second-order Taylor expansion is

$$dV = V_t\,dt + V_S\,dS + \frac{1}{2}V_{SS}(dS)^2 + V_{St}\,dS\,dt + \frac{1}{2}V_{tt}(dt)^2 + \cdots.$$

The underlying asset follows

$$dS = \mu S\,dt + \sigma S\,dW.$$

Therefore,

$$(dS)^2 = (\mu S\,dt + \sigma S\,dW)^2.$$

Expanding,

$$(dS)^2 = \mu^2 S^2 (dt)^2 + 2\mu\sigma S^2\,dt\,dW + \sigma^2 S^2 (dW)^2.$$

Using

$$(dt)^2 = 0,$$

$$dt\,dW = 0,$$

and

$$(dW)^2 = dt,$$

we obtain

$$\boxed{(dS)^2 = \sigma^2 S^2\,dt.}$$

16.6 Task 2: Derive the Dynamics of the Derivative

Substituting into the Taylor expansion gives

$$dV = V_t\,dt + V_S\,dS + \frac{1}{2}V_{SS}\sigma^2 S^2\,dt.$$

Now substitute

$$dS = \mu S\,dt + \sigma S\,dW.$$

Hence,

$$\boxed{dV = \left[V_t + \mu S V_S + \frac{1}{2}\sigma^2 S^2 V_{SS}\right]dt + \sigma S V_S\,dW.}$$

This equation should be derived line by line.

It should not be introduced merely as a formula to be used.

16.7 Task 3: Numerical Illustration of Itô's Lemma

The quadratic-variation experiment completed in Week 1 should now be connected directly with Itô's lemma.

Choose a simple function, for example,

$$f(W) = W^2.$$

Itô's lemma gives

$$df = 2W\,dW + dt.$$

Thus,

$$\boxed{d(W^2) = 2W\,dW + dt.}$$

Ordinary calculus would incorrectly suggest only

$$d(W^2) = 2W\,dW.$$

A numerical experiment should demonstrate the importance of the additional dt term.

16.8 Task 4: A Second Itô Experiment

Use

$$f(S) = \ln S.$$

For

$$dS = \mu S\,dt + \sigma S\,dW,$$

Itô's lemma gives

$$d(\ln S) = \frac{1}{S}dS - \frac{1}{2S^2}(dS)^2.$$

Therefore,

$$\boxed{d(\ln S) = \left(\mu - \frac{1}{2}\sigma^2\right)dt + \sigma\,dW.}$$

Integrating gives

$$\boxed{S_T = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)T + \sigma W_T\right].}$$

This provides the mathematical derivation of the GBM solution that was used computationally in Week 1.

16.9 Task 5: Construct the Delta-Hedged Portfolio

Consider a portfolio containing

one derivative

and

$$-\Delta$$

units of the underlying asset.

Define

$$\boxed{\Pi = V - \Delta S.}$$

Over an infinitesimal interval,

$$d\Pi = dV - \Delta\, dS.$$

Substitute the expressions for dV and dS.

16.10 Task 6: Choose the Hedge Ratio

Choose

$$\boxed{\Delta = V_S.}$$

Then

$$d\Pi = dV - V_S\, dS.$$

The stochastic term

$$\sigma S V_S\, dW$$

in dV is cancelled by the corresponding stochastic term in

$$V_S\, dS.$$

The result is

$$\boxed{d\Pi = \left[V_t + \frac{1}{2}\sigma^2 S^2 V_{SS}\right] dt.}$$

The portfolio has therefore become locally riskless.

16.11 The Central Question of Week 2

At this point the following question should be answered explicitly:

$$\boxed{\text{Where did the physical drift } \mu \text{ go?}}$$

It disappears because the stochastic exposure is eliminated through the hedge

$$\Delta = V_S.$$

This is one of the most important conceptual points in the entire Black–Scholes derivation.

16.12 Task 7: Apply the No-Arbitrage Argument

Since the hedged portfolio is locally riskless, it must earn the risk-free rate.

Therefore,

$$d\Pi = r\Pi\, dt.$$

Using

$$\Pi = V - SV_S,$$

we obtain

$$\left[V_t + \frac{1}{2}\sigma^2 S^2 V_{SS}\right] dt = r\,(V - SV_S)\, dt.$$

Rearranging gives

$$\boxed{V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0.}$$

This is the Black–Scholes partial differential equation.

16.13 Interpretation of the Black–Scholes PDE

Each term should be interpreted rather than memorized.

V_t	:	time effect,
$\frac{1}{2}\sigma^2 S^2 V_{SS}$	:	curvature–volatility effect,
rSV_S	:	underlying-price sensitivity under risk-neutral growth,
$-rV$	:	discounting / financing effect.

16.14 Task 8: Introduce the Terminal Payoff

For a European call,

$$\boxed{V(S,T) = \max(S - K, 0).}$$

For a European put,

$$\boxed{V(S,T) = \max(K - S, 0).}$$

The Black–Scholes problem is therefore not merely the PDE

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0,$$

but the PDE together with the appropriate terminal condition.

16.15 Task 9: Implement the European Call Formula

For a European call,

$$\boxed{C = SN(d_1) - Ke^{-rT}N(d_2),}$$

where

$$d_1 = \frac{\ln(S/K) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}$$

and

$$d_2 = d_1 - \sigma\sqrt{T}.$$

16.16 Task 10: Implement the European Put Formula

The corresponding European put price is

$$\boxed{P = Ke^{-rT}N(-d_2) - SN(-d_1).}$$

16.17 Task 11: Put–Call Parity

Verify numerically that

$$\boxed{C - P = S - Ke^{-rT}.}$$

Define the parity error

$$\varepsilon_{\text{PCP}} = C - P - S + Ke^{-rT}.$$

For a correct implementation,

$$\boxed{|\varepsilon_{\text{PCP}}| \approx 0.}$$

The verification should be carried out over a grid of

$$S, \qquad K, \qquad T, \qquad r, \qquad \sigma.$$

16.18 Task 12: Understand the Black–Scholes Inputs

The option price should now be varied systematically with respect to

$$S, \qquad K, \qquad T, \qquad r, \qquad \sigma.$$

Only one variable should initially be changed at a time.

16.18.1 Effect of the Underlying Price

Plot

$$C(S)$$

for fixed

$$K, \quad T, \quad r, \quad \sigma.$$

Verify that

$$\boxed{\frac{\partial C}{\partial S} > 0.}$$

16.18.2 Effect of Strike

Plot

$$C(K).$$

Verify numerically that increasing K decreases the call value.

16.18.3 Effect of Volatility

Plot

$$C(\sigma)$$

for several levels of moneyness.

Verify that

$$\boxed{\frac{\partial C}{\partial \sigma} > 0.}$$

16.18.4 Effect of Time to Maturity

Plot

$$C(T)$$

for suitable European calls and interpret the result.

16.18.5 Moneyness

Introduce

$$m = \ln\left(\frac{S}{K}\right).$$

Identify the regions

$m < 0$	$m \approx 0$	$m > 0$
OTM	ATM	ITM.

16.19 Task 13: Connect the Derivation Back to Week 1

The derivation should now be examined against the empirical findings already obtained.

Prepare a table of the form

Classical Model Assumption	**Empirical Observation**
Continuous diffusion	Examine extreme market movements
Constant volatility	Volatility clustering already observed
Gaussian log-return innovation	Strong skewness and excess kurtosis observed
Independent increments	Return magnitudes exhibit persistent dependence
Lognormal asset-price dynamics	Useful benchmark, but incomplete description of market behaviour

The purpose is not to reject Black–Scholes.

The purpose is to understand exactly what mathematical structure is being assumed before the model is used as the benchmark for the later AI/ML framework.

16.20 Week 2 Expected Deliverables

By the end of Week 2, the following should be available:

`week03_ito_black_scholes.ipynb`
`black_scholes.py`
Itô Numerical Verification
Black–Scholes Derivation
Call and Put Pricing Functions
Put–Call Parity Tests
Parameter-Sensitivity Plots.

16.21 Mandatory Figures

The Week 2 report should contain at least the following figures:

1. Numerical illustration of
$$d(W^2) = 2W\,dW + dt.$$
2. Call price versus underlying price,
$$C(S).$$
3. Call price versus volatility,
$$C(\sigma).$$
4. Call price versus maturity,
$$C(T).$$
5. Call price versus moneyness,
$$C\left(\ln(S/K)\right).$$

16.22 Questions to be Answered at the Week 2 Review

The following questions should be answerable without referring to the Python code.

1. Why is ordinary Taylor calculus insufficient for Brownian motion?
2. What is meant by
$$(dW)^2 = dt?$$
3. Why does the second-order term survive in Itô's lemma?
4. How does Itô's lemma produce the term
$$-\frac{1}{2}\sigma^2$$
in the logarithm of a GBM process?
5. What is the difference between the empirical Taylor/Jensen correction examined previously and the Itô correction?
6. What is the financial meaning of
$$\Delta = V_S?$$
7. Which stochastic term is eliminated by delta hedging?

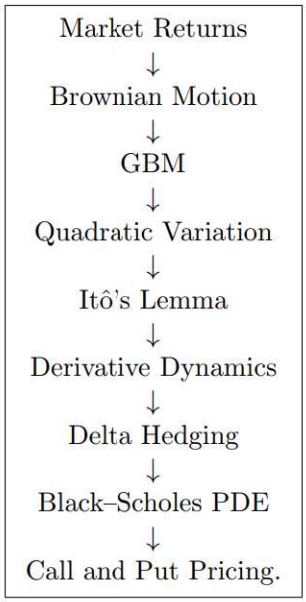
8. Why does μ disappear from the Black–Scholes PDE?
9. Why must a locally riskless portfolio earn the risk-free rate?
10. Why is a terminal condition required in addition to the PDE?
11. What is the difference between an ITM, ATM and OTM option?
12. What does put–call parity test in the implementation?

16.23 Week 2 Progress Checkpoint

Component	**Status**
Understand quadratic variation and stochastic differential rules	☐
Derive Itô's lemma application to $V(S,t)$	☐
Numerically verify the Itô correction	☐
Derive GBM solution using $f(S) = \ln S$	☐
Construct the delta-hedged portfolio	☐
Demonstrate cancellation of dW	☐
Derive the Black–Scholes PDE	☐
Explain disappearance of μ	☐
Implement European call and put prices	☐
Verify put–call parity	☐
Complete parameter-sensitivity experiments	☐
Relate model assumptions to Week 1 empirical findings	☐

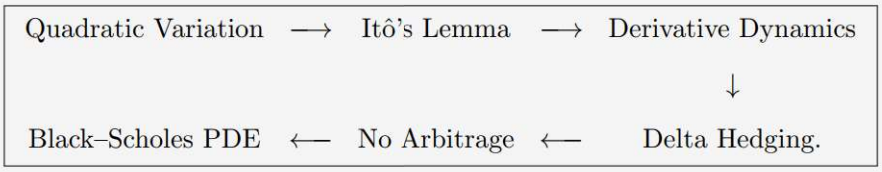
16.24 Position Expected at the End of Week 2

At the end of Week 2, the project should have progressed to



Project Milestone

The Week 2 milestone is reached only when the Black–Scholes equation can be derived and explained without treating it as a memorized formula.
The essential connection is



Once this connection is established, Week 3 can proceed naturally to Greeks, sensitivity analysis and implied volatility.

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